MIL-B-48600A (AR)
30 OCTOBER 1979
SUPERSEDING
MIL-B-48600 (MU)
16 July 1976

# MILITARY SPECIFICATION

# BASE PLUG ASSEMBLY

This specification if approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope.-This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of a base plug seal to be used in the 105MM, APDS-T and TPDS-T ammunition.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents.-The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-C-60539 - Coatings, Anodic, Conventional, for Aluminum and Aluminum Alloys.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1315

**Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.**

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

DRAWINGS

11820563 - Base Plug Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM Methods - American Society for Testing Materials

E8 - Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa 19103).

## 3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Heat treatment or stress relief.-Heat treatment or stress relief applied to any component part shall be uniform throughout and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing.

3.3 Mechanical properties.-Mechanical properties shall comply with the requirements of the applicable drawing or the applicable material specification throughout the component.

3.4 Surface preparation and application of protective finish.-Where cleaning and surface pretreatment are required by component details, the process shall apply to all indicated surfaces regardless of contour configuration. The protective finish shall be as specified on the applicable drawings and shall be uniformly applied to the surfaces designated by the applicable drawings. Salt spray and other required testing shall be in accordance with the applicable drawings and specifications.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.5 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship.-The requirements for workmanship are as shown on the applicable component drawing, in the applicable specifications and the following:

3.6.1 Material defects.-Soundness of material shall be as required by the applicable specifications and drawings. In addition, all components shall be free from cracks, splits, cold shuts, inclusions, porosity and any other similar defects.

3.6.2 Protective coating.-Protective coating shall be uniform and undamaged.

3.6.3 Surface finish.-Surface shall be in accordance with the applicable drawing and the specification referenced thereon.

3.6.4 Burr.-No component shall have a burr.

3.6.5 Foreign matter.-All parts, sub-assemblies and assemblies shall be free of chips, dirt, grease, kerosene, rust, corrosion or other foreign matter.

3.6.6 Discoloration.-Discoloration is not allowed.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection



Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the currentversion before use.

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the items in sample quantities as indicated below.

| Part Description | Drawing | Quantity |
|---|---|---|
| Base Plug | 11820562 | 10* |
| Insert, Base Plug | 11820534 | 10* |
| Base Plug Assembly | 11820563 | 10* |

5 ends of 5 different rods used in the manufacture of the base plugs, with a minimum length of one foot.
5 ends of 5 different rods used in the manufacture of base plug inerts, with a minimum length of one foot.

*Half of the samples shall be submitted without protective coating and the other half with protective coating.

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE I - FIRST ARTICLE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|
| | Base Plug Assembly | | | See Below |
| | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | I Base Plug - Dwg. | | | | |
| | a. Examination for defects | 10 | | | 4.4.2.1 |
| | b. Salt spray test | 5 | | | See Note |
| | c. Tension test | 2 | | | 4.5.2 |
| | II Insert, Base Plug - Dwg. | | | | |
| | a. Examination for defects | 10 | | | 4.4.2.2 |
| | III Base Plug Assembly - Dwg. | | | | |
| | a. Examination for defects | 10 | | | 4.4.2.3 |

NOTES: The samples shall be submitted for the salt spray test in accordance with MIL-A-60539 for 240 hrs. If any sample fails this test the first article shall be rejected.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plan.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Tool control dimensions.-Dimensions marked tool control shall be gaged during the First Article testing and twice in every 8 hours shift of regular production and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, mesurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor when drawing requirements are not met.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48600A (AR)

| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1 | Base Plug | | 11820562 |
| | | | NEXT HIGHER ASSEMBLY |
| | | | 11820563 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Protective coating inadequate | See 4.5.1 | | 3.4 | 4.5.1 |
| 102 | Salt spray test | 5 | | 3.4 | 4.5.1.1 |
| 103 | Tension test | See Note | | 3.3 | 4.5.2 |
| 104 | True position of large inside diameter with outside diameter | | 0.40% | 3.1 | Gage |
| 105 | True position of small inside diameter with outside diameter | | 0.40% | 3.1 | Gage |
| 106 | Runout of radius with large outside diameter | | 0.40% | 3.1 | Gage |
| 107 | Diameter, large outside | | 0.40% | 3.1 | Gage |
| 108 | Diameter, large inside | | 0.40% | 3.1 | Gage |
| 109 | Diameter, small inside | | 0.40% | 3.1 | Gage |
| 110 | Radii incorrect | See 4.4.2.b | | 3.1 | 4.4.2.b |
| Minor | | | | | |
| 201 | Length total | | 0.65% | 3.1 | Gage |
| 202 | Length to shoulder | | 0.65% | 3.1 | Gage |
| 203 | Length to counterbore | | 0.65% | 3.1 | Gage |
| 204 | Distance from rear face to inside spherical radius | | 0.65% | 3.1 | Gage |
| 205 | Diameter of head | | 0.65% | 3.1 | Gage |
| 206 | Diameter of flat | | 0.65% | 3.1 | Gage |
| 207 | Radii or chamfers, missing or inocrrect | | 0.65% | 3.1 | Visual |
| 208 | Evidence of poor workmanship | | 0.65% | 3.6 | Visual |

NOTES: Heat treated bar lots containing 500 feet or less shall have one specimen removed from each of two bars and subjected to this test. Bar lots containing more than 500 feet shall have one additional specimen removed and tested for each additional 1,000 feet or fraction thereof. If any test specimen fails the test, the lot shall be rejected.



7
Source: https://assist.dla.mil -- Downloaded: 2026/10-26T03:56Z
Check the source to verify that this is the current version before use.

# CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.2 | Insert, Base Plug | | | | 11820534 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 11820563 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | True position of inside diameter with outside diameter of body | | 0.40% | 3.1 | Gage |
| 102 | Diameter of body, outside | | 0.40% | 3.1 | Gage |
| 103 | Diameter of body, inside | | 0.40% | 3.1 | Gage |
| 104 | Protective coating inadequate | See | 4.5.1 | 3.4 | 4.5.1 |
| Minor | | | | | |
| 201 | Diameter of shoulder | | 0.65% | 3.1 | Gage |
| 202 | Length of shoulder | | 0.65% | 3.1 | Gage |
| 203 | Length, total | | 0.65% | 3.1 | Gage |
| 204 | Chamber, missing or incorrect | | 0.65% | 3.1 | Visual |
| 205 | Evidence of poor workmanship | | 0.65% | 3.6 | Visual |

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48600A (AR)

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.3 | Base Plug Assembly | 1 OF 1 | 11820563 |
| | | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Loose assembly | | 0.40% | 3.1 | Visual/Manual |
| 102 | Base plug insert improperly seated | | 0.40% | 3.1 | Gage |
| Minor | | | | | |
| 201 | Protective coating damaged | | 0.65% | 3.6.2 | Visual |
| 202 | Evidence of poor workmanship | | 0.65% | 3.6 | Visual |

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.4.3 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test method and procedures

4.5.1 Protective coating.-Procedures for testing, the cleaning of components and the application of protective coatings shall be in accordance with the criteria for testing as specified on the applicable component drawings and in the specification reference thereon. If a salt spray is required on the applicable specification, the salt spray test shall be accomplished on components or assemblies which have been manufactured in accordance with the drawing requirements.

4.5.1.1 Salt spray test - base plug.-The salt spray test shall be accomplished in accordance with MIL-C-60539. During regular production the samples shall be exposed for 120 hours, beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements.

After three consecutive lots have been accepted, the samples of all subsequent lots shall be exposed for 72 hours. Should any lot fail the applicable requirements, the salt spray samples of the following lot shall be exposed for 120 hours until three consecutive lots have been accepted.

4.5.2 Tension test - base plug.-One tensile test specimen shall be taken longitudinally from the center of each bar. The specimen shall conform to the largest obtainable round specimen described by ASTM E-8. Yield strength shall be determined by the offset method utilizing the procedures prescribed by ASTM E-8.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking

5.1.1 Level A - Military package.-NA

5.1.2 Level B - Minimum military package.-Preservation, packaging, packing and marking shall omply with the applicable drawing, specification, military specification MIL-STD-1169 and this specification.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

5.2 Packing.-Base plug assemblies shall be packed in a manner which will insure acceptance by common carrier and will afford protection against physical damage during shipment from the supply source to the first receiving activity for immediate use. Shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Marking.-In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Shipping.-When components from more than one lot are shipped as a car load, each lot must be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Ordering data.-See MIL-A-48078.

a. Data Cards - Data cards shall be prepared for each lot in accordance with MIL-STD-1167.

b. Delivery of data cards to the technical agency.-The following address is to be added to those listed in the contract (Form DD 1423) for the delivery of one copy, to the technical agency of the Ammunition Data Cards:

Commander, US Army Armament
Research and Development Command
ATTN: DRDAR-QAR-E
Dover, New Jersey 07801

6.3 Submission of inspection equipment designs for approval.-See MIL-A-48078. Submit equipment designs, as required to:

Commander
US ARmy Research and Development Command
ATTN: DRDAR-QAR-I
Dover, NJ 07801

Custodian:
ARMY-AR

Preparing Activity:
ARMY-AR
Project No. 1315-A236



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

**NOTE**: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-48600A Base Plug Assembly

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

| SUBMITTED BY *(Printed or typed name and address — Optional)* | TELEPHONE NO. |
|---|---|
| | DATE |

**DD FORM 1 OCT 76 1426** EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.